ResearchGate

See discussions, stats, and author profiles for this publication at: https://www.researchgate.net/publication/352018276

Lung Cancer

Chapter · June 2021

CITATIONS
0

READS
4,234

2 authors:

Major Mirza Qaiser Baig
J.K Cancer Institute Kanpur
502 PUBLICATIONS **14** CITATIONS

SEE PROFILE

Dev Kumar Yadav
Motilal Nehru Medical College
33 PUBLICATIONS **5** CITATIONS

SEE PROFILE

All content following this page was uploaded by Dev Kumar Yadav on 01 June 2021.

The user has requested enhancement of the downloaded file.

Chapter

35

Lung Cancer

INTRODUCTION

Lung cancer is a disease of uncontrolled cell growth in tissues of the lung. This growth may lead to metastasis, which is the invasion of adjacent tissue and infiltration beyond the lungs. The vast majority of primary lung cancers are **carcinomas of the lung**, derived from epithelial cells. Lung cancer, the most common cause of cancer-related death in men and women, is responsible for 1.3 million deaths worldwide annually, as of 2004 The most common symptoms are shortness of breath, coughing (including coughing up blood), and weight loss. The main types of lung cancer are small cell lung carcinoma and non-small cell lung carcinoma. This distinction is important, because the treatment varies; non-small cell lung carcinoma (NSCLC) is sometimes treated with surgery, while small cell lung carcinoma (SCLC) usually responds better to chemotherapy and radiation. The most common cause of lung cancer is long-term exposure to tobacco smoke The occurrence of lung cancer in nonsmokers, who account for as many as 15% of cases, is often attributed to a combination of genetic factors, radon gas, asbestos, and air pollution, including secondhand smoke.

Lung cancer may be seen on chest radiograph and computed tomography (CT scan). The diagnosis is confirmed

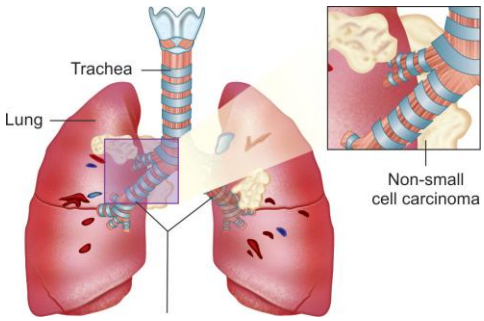

Fig. 35.1:NON SMALL CELL LUNG CANCER

with a biopsy. This is usually performed by bronchoscopy or CT-guided biopsy. Treatment and prognosis depend upon the histological type of cancer, the stage (degree of spread), and the patient's performance status. Possible treatments include surgery, chemotherapy, and radiotherapy. Depending on the stage and

treatment, the five-year survival rate is 14%.

Classification

The vast majority of lung cancers are carcinomas—malignancies that arise from epithelial cells. There are two main types of lung carcinoma, categorized by the size and appearance of the malignant cells seen by a histopathologist under a microscope: nonsmall cell (80.4%) and small-cell (16.8%) lung carcinoma. This classification, based on histological criteria, has important implications for clinical management and prognosis of the disease.

Non-small Cell Lung Carcinoma (NSCLC)

The non-small cell lung carcinomas are grouped together because their prognosis and management are similar. There are three main sub-types: squamous cell lung carcinoma, adenocarcinoma, and large cell lung carcinoma Accounting for 25% of lung cancers,] squamous cell lung carcinoma usually starts near a central bronchus. A hollow cavity and associated necrosis are commonly found at the center of the tumor. Well-differentiated squamous cell lung cancers often grow more slowly than other cancer types. Adenocarcinoma accounts for 40% of lung cancers.] It usually originates in peripheral lung tissue. Most cases of adenocarcinoma are associated with smoking; however, among people who have never smoked (“nonsmokers”), adenocarcinoma is the most common form of lung cancer. A subtype of adenocarcinoma, the bronchioloalveolar carcinoma, is more common in female nonsmokers, and may have different responses to treatment.]

Small Cell Lung Carcinoma (SCLC)

Small cell lung carcinoma (microscopic view of a core needle biopsy).

Small cell lung carcinoma is less common. Also called oat cell cancer, it tends to arise in the larger airways (primary and secondary bronchi) and grows rapidly, becoming quite large.] The small cells contains dense neurosecretory granules (vesicles containing neuroendocrine hormones), which give this an endocrine/paraneoplastic syndrome

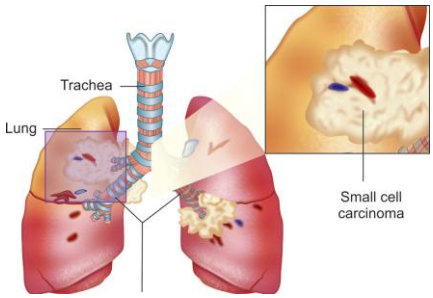

Fig. 35.2: SMALL CELL LUNG CANCER

association.] While initially more sensitive to chemotherapy, it ultimately carries a worse prognosis and is often metastatic at presentation. Small cell lung cancers are divided into limited stage and extensive stage disease. This type of lung cancer is strongly associated with smoking.

Others

In infants and children, the most common primary lung cancers are pleuropulmonary blastoma and carcinoid tumor.

Secondary Cancers

The lung is a common place for metastasis from tumors in other parts of the body. These secondary cancers are identified by the site of origin; thus, a breast cancer metastasis to the lung is still known as breast cancer. They often have a characteristic round appearance on chest radiograph. In children, the majority of lung cancers are secondary.

Primary lung cancers themselves most commonly metastasize to the adrenal glands, liver, brain, and bone.

Staging

Lung cancer staging is an assessment of the degree of spread of the cancer from its original source. It is an important factor affecting the prognosis and potential treatment of lung cancer. Non-small cell lung carcinoma is staged from IA ("one A"; best prognosis) to IV ("four"; worst prognosis). Small cell lung carcinoma is classified as limited stage if it is confined to one half of the chest and within the scope of a single radiotherapy field; otherwise, it is extensive stage.

Signs and Symptoms

Symptoms that suggest lung cancer include: dyspnea (shortness of breath), hemoptysis (coughing up blood), chronic coughing or

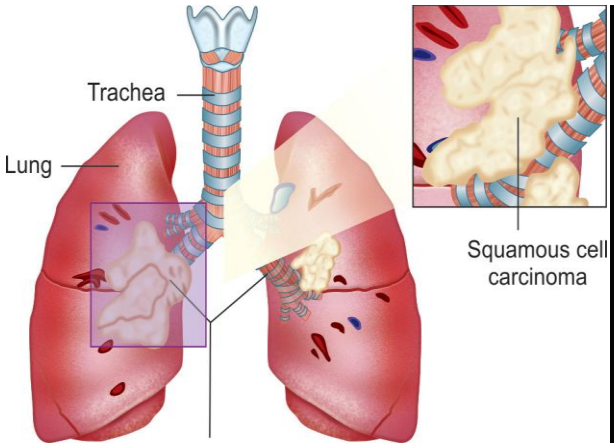

Fig. 35.3: SQUMOUS CELL CARCINOMA LUNG

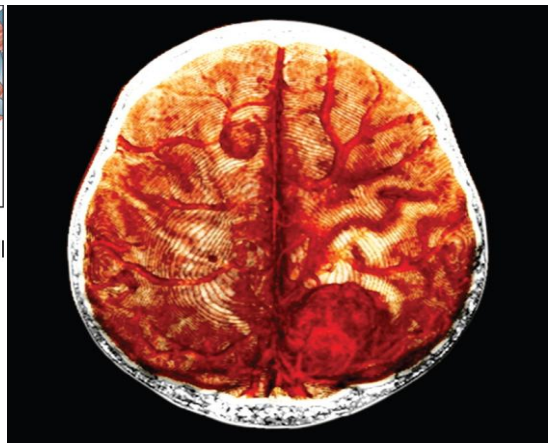

Fig. 35.4: BRAIN METASTASIS IN LUNG CANCER

change in regular coughing pattern, wheezing, chest pain or pain in the abdomen, cachexia (weight loss), fatigue, and loss of appetite ,dysphonia (hoarse voice) clubbing of the fingernails (uncommon) dysphagia (difficulty swallowing). If the cancer grows in the airway, it may obstruct airflow, causing breathing difficulties. This can lead to accumulation of secretions behind the blockage, predisposing the patient to pneumonia. Many lung cancers have a rich blood supply. The surface of the cancer may be fragile, leading to bleeding from the cancer into the airway. This blood may subsequently be coughed up. Depending on the type of tumor, so-called paraneoplastic phenomena may initially attract attention to the disease. In lung cancer, these phenomena may include Lambert-Eaton myasthenic syndrome (muscle weakness due to autoantibodies), hypercalcemia, or syndrome of inappropriate antidiuretic hormone (SIADH). Tumors in the top (apex) of the lung, known as Pancoast tumors, may invade the local part of the sympathetic nervous system, leading to changed sweating patterns and eye muscle problems (a combination known as Horner's syndrome) as well as muscle weakness in the hands due to invasion of the brachial plexus. Many of the symptoms of lung cancer (bone pain, fever, and weight loss) are nonspecific; in the elderly, these may be attributed to co morbid illness. In many patients, the cancer has already spread beyond the original site by the time they have symptoms and seek medical attention. Common sites of metastasis include the brain, bone, adrenal glands, contralateral (opposite) lung, liver, pericardium, and kidneys. About 10% of people with lung cancer do not have symptoms at diagnosis; these cancers are incidentally found on routine chest radiograph.

Causes

The main causes of any cancer include carcinogens (such as those in tobacco smoke), ionizing radiation, and viral infection. This exposure causes cumulative changes to the DNA in the tissue lining the bronchi of the lungs (the bronchial epithelium). As more tissue becomes damaged, eventually a cancer develops.

Smoking

Following an increase in smoking in the population, the rate of lung cancer death increases. Smoking, particularly of cigarettes, is by far the main contributor to lung cancer. Across the developed world, almost 90% of lung cancer deaths are caused by smoking. In the United States, smoking is estimated to account for 87% of lung cancer cases (90% in men and 85% in women

Among male smokers, the lifetime risk of developing lung cancer is 17.2%; among female smokers, the risk is 11.6%. This risk is significantly lower in nonsmokers: 1.3% in men and 1.4% in women. Cigarette smoke contains over 60 known carcinogens,] including radioisotopes from the radon decay sequence, nitrosamine, and benzopyrene. Additionally, nicotine appears to depress the immune response to malignant growths in exposed tissue. The time a person smokes (as well as rate of smoking) increases the person's chance of developing lung cancer. If a person stops smoking, this chance steadily decreases as damage to the lungs is repaired and contaminant particles are gradually removed.[39] In addition, there is evidence that lung cancer in never-smokers has a better prognosis than in smokers, and that patients who smoke at the time of diagnosis have shorter survival times than those who have quit. Passive smoking—the inhalation of smoke from another's smoking—is a cause of lung cancer in nonsmokers. A passive smoker can be classified as someone living or working with a smoker as well. Studies have consistently shown a significant increase in relative risk among those exposed to passive

Ch-35.indd 270 11-06-2015 10:26:11

smoke. Recent investigation of sidestream smoke suggests that it is more dangerous than direct smoke inhalation.

Roughly ten-fifteen percent of lung cancer patients have never smoked. That means between 20,000 to 30,000 never-smokers are diagnosed with lung cancer in the United States each year. Because of the five-year survival rate, each year in the U.S. more nonsmokers die of lung cancer than do patients of leukemia, ovarian cancer, or AIDS.

Radon gas, Radon is a colorless and odorless gas generated by the breakdown of radioactive radium, which in turn is the decay product of uranium, found in the Earth's crust. The radiation decay products ionize genetic material, causing mutations that sometimes turn cancerous. Radon exposure is the second major cause of lung cancer, after smoking, Radon gas levels vary by locality and the composition of the underlying soil and rocks. For example, in areas such as Cornwall in the UK (which has granite as substrata), radon gas is a major problem, and buildings have to be force-ventilated with fans to lower radon gas concentrations. The United States Environmental Protection Agency (EPA) estimates that one in 15 homes in the U.S. has radon levels above the recommended guideline of 4 picocuries per liter (pCi/L) (148

Bq/m³). Iowa has the highest average radon concentration in the United States; studies performed there have demonstrated a 50% increased lung cancer risk, with prolonged radon exposure above the EPA's action level of 4 pCi/L.

Asbestos

Asbestos can cause a variety of lung diseases, including lung cancer. There is a synergistic effect between tobacco smoking and asbestos in the formation of lung cancer. In the UK, asbestos accounts for 2–3% of male lung cancer deaths.] Asbestos can also cause cancer of the pleura, called mesothelioma (which is different from lung cancer).

Viruses

Viruses are known to cause lung cancer in animals, and recent evidence suggests similar potential in humans. Implicated viruses include human papillomavirus, JC virus,] simian virus 40 (SV40), BK virus, and cytomegalovirus. These viruses may affect the cell cycle and inhibit apoptosis, allowing uncontrolled cell division.

Particulate Matter: Studies of the American Cancer Society cohort directly link the exposure to particulate matter with lung cancer. For example, if the concentration of particles in the air increases by only 1%, the risk of developing a lung cancer increases by 14%. Further, it has been established that particle size matters, as ultrafine particles penetrate further into the lungs.

Pathogenesis

Similar to many other cancers, lung cancer is initiated by activation of oncogenes or inactivation of tumor suppressor genes.

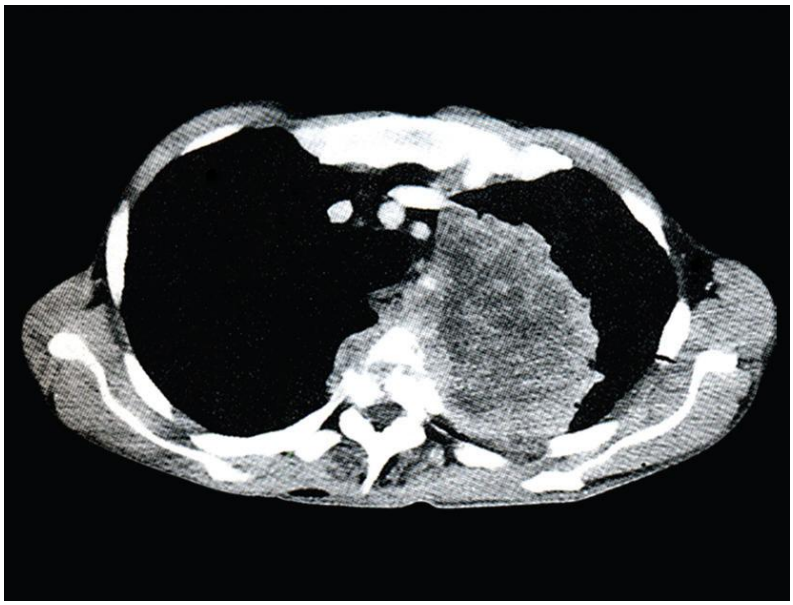

Fig. 35.5: CECT THORAX SHOWING LEFT LUNG MASS

Oncogenes are genes that are believed to make people more susceptible to cancer. Proto-oncogenes are believed to turn into oncogenes when exposed to particular carcinogens. Mutations in the *K-ras* protooncogene are responsible for 10–30% of lung adenocarcinomas. The epidermal growth factor receptor (EGFR) regulates cell proliferation, apoptosis, angiogenesis, and tumor invasion.[] Mutations and amplification of EGFR are common in non-small cell lung cancer and provide the basis for treatment with EGFR-inhibitors. Her2/neu is affected less frequently. Chromosomal damage can lead to loss of heterozygosity. This can cause inactivation of tumor suppressor genes. Damage to chromosomes 3p, 5q, 13q, and 17p are particularly common in small cell lung carcinoma. The *p53* tumor suppressor gene, located on chromosome 17p, is affected in 60-75% of cases. Other genes that are often mutated or amplified are *c-MET*, *NKX2-1*, *LKB1*, *PIK3CA*, and *BRAF*.

Several genetic polymorphisms are associated with lung cancer. These include polymorphisms in genes coding for interleukin-1, cytochrome P450, apoptosis promoters such as caspase-8,, and DNA repair molecules such as XRCC1.[x] People with these polymorphisms are more likely to develop lung cancer after exposure to carcinogens.

A recent study suggested that the MDM2 309G allele, is a low-penetrant risk factor for developing lung cancer in Asians.

Diagnosis

Performing a chest radiograph is the first step if a patient reports symptoms that may suggest lung cancer. This may reveal an obvious mass, widening of the mediastinum (suggestive of spread to lymph nodes there), atelectasis (collapse), consolidation (pneumonia), or pleural effusion. If there are no radiographic findings but the suspicion is high (such as a heavy smoker with blood-stained sputum), bronchoscopy and/or a CT scan may provide the necessary information. Bronchoscopy or CT-guided biopsy is often used to identify the tumor type.

Abnormal findings in cells (“atypia”) in sputum are associated with an increased risk of lung cancer. Sputum cytologic examination combined with other screening examinations may have a role in the early detection of lung cancer.

The differential diagnosis for patients who present with abnormalities on chest radiograph includes lung cancer as well as nonmalignant diseases. These include infectious causes such as tuberculosis or pneumonia, or inflammatory conditions such as sarcoidosis. These diseases can result

in mediastinal lymphadenopathy or lung nodules, and sometimes mimic lung cancers. Lung cancer can also be an incidental finding: a solitary pulmonary nodule (also called a coin lesion) on a chest radiograph or CT scan taken for an unrelated reason.

Prevention

Prevention is the most cost-effective means of fighting lung cancer. While in most countries industrial and domestic carcinogens have been identified and banned, tobacco smoking is still widespread. Eliminating tobacco smoking is a primary goal in the prevention of lung cancer, and smoking cessation is an important preventive tool in this process Most importantly, are prevention programs that target the young. In 1998 the Master Settlement Agreement entitled 46 states in the USA to an annual payout from the tobacco companies. Between the settlement money and tobacco taxes, each state's public health department funds their prevention programs, although none of the states are living up to the Center for Disease Control's recommended amount by spending 15 percent of tobacco taxes and settlement revenues on these prevention efforts. Policy interventions to decrease

Ch-35.indd 272 11-06-2015 10:26:11

passive smoking in public areas such as restaurants and workplaces have become more common in many Western countries, with California taking a lead in banning smoking in public establishments in 1998. Ireland played a similar role in Europe in 2004, followed by Italy and Norway in 2005, Scotland as well as several others in 2006, England in 2007, France in 2008 and Turkey in 2009. New Zealand has banned smoking in public places as of 2004. The state of Bhutan has had a complete smoking ban since 2005. In many countries, pressure groups are campaigning for similar bans. In 2007, Chandigarh became the first city in India to become smoke-free city of India introduced a total ban on smoking at public places on Oct 2 2008.

Arguments cited against such bans are criminalisation of smoking, increased risk of smuggling, and the risk that such a ban cannot be enforced. The long-term use of supplemental multivitamins—such as vitamin C, vitamin E, and folate—does not reduce the risk of lung cancer. Indeed long-term intake of high doses of vitamin E supplements may even increase the risk of lung cancer.

The World Health Organization has called for governments to institute a total ban on tobacco advertising to prevent young people from taking up smoking. They assess that such bans have reduced tobacco consumption by 16% where already instituted.

Screening

Lung Cancer Screening

Screening refers to the use of medical tests to detect disease in asymptomatic people. Possible screening tests for lung cancer include chest radiograph or computed tomography (CT).

Treatment of Lung Cancer

Treatment for lung cancer depends on the cancer's specific cell type, how far it has spread, and the patient's performance status. Common treatments include surgery, chemotherapy, and radiation therapy.

Surgery

Gross appearance of the cut surface of a pneumonectomy specimen containing a lung cancer, here a squamous cell carcinoma (the whitish tumor near the bronchi).

If investigations confirm lung cancer, CT scan and often positron emission tomography (PET) are used to determine whether the disease is localized and amenable to surgery or whether it has spread to the point where it cannot be cured surgically.

Blood tests and spirometry (lung function testing) are also necessary to assess whether the patient is well enough to be operated on. If spirometry reveals poor respiratory reserve (often due to chronic obstructive pulmonary disease), surgery may be contraindicated., Surgery itself has an operative death rate of about 4.4%, depending on the patient's lung function and other risk factors. Surgery is usually only an option in non-small cell lung carcinoma limited to one lung, up to stage IIIA. This is assessed with medical imaging (computed tomography, positron emission tomography). A sufficient preoperative respiratory reserve must be present to allow adequate lung function after the tissue is removed., Procedures include wedge resection (removal of part of a lobe), segmentectomy (removal of an anatomic division of a particular lobe of the lung), lobectomy (one lobe), bilobectomy (two lobes), or pneumonectomy (whole lung). In patients with adequate respiratory reserve, lobectomy is the preferred option, as this minimizes the chance of local recurrence. If the patient does not have enough functional lung for this, wedge resection may be performed. Radioactive iodine brachytherapy at the margins of wedge excision may reduce recurrence to that of lobectomy. Video-assisted thoracoscopic surgery and VATS lobectomy have allowed for minimally invasive approaches to lung cancer surgery that may

have the advantages of quicker recovery, shorter hospital stay and diminished hospital costs

Chemotherapy

Small cell lung carcinoma is treated primarily with chemotherapy and radiation, as surgery has no demonstrable influence on survival. Primary chemotherapy is also given in metastatic non-small cell lung carcinoma. The combination regimen depends on the tumor type. Non-small cell lung carcinoma is often treated with cisplatin or carboplatin, in combination with gemcitabine, paclitaxel, docetaxel, etoposide, or vinorelbine. In small cell lung carcinoma, cisplatin and etoposide are most commonly used. Combinations with carboplatin, gemcitabine, paclitaxel, vinorelbine, topotecan, and irinotecan are also used. in extensive-stage small-cell lung cancer celecoxib may safely be combined with etoposide, this combination showed improve outcomes.

Adjuvant Chemotherapy for NSCLC

Adjuvant chemotherapy refers to the use of chemotherapy after surgery to improve the outcome. During surgery, samples are taken from the lymph nodes. If these samples contain cancer, the patient has stage II or III disease. In this situation, adjuvant chemotherapy may improve survival by up to 15%. Standard practice is to offer platinum-based chemotherapy (including either cisplatin or carboplatin). Adjuvant chemotherapy for patients with stage IB cancer is controversial, as clinical trials have not clearly demonstrated a survival benefit Trials of preoperative chemotherapy (neoadjuvant chemotherapy) in resectable non-small cell lung carcinoma have been inconclusive.

Radiotherapy

Radiotherapy is often given together with chemotherapy, and may be used with curative intent in patients with non-small cell lung carcinoma who are not eligible for surgery. This form of high intensity radiotherapy is called radical radiotherapy.[96] A refinement of this technique is continuous hyperfractionated accelerated radiotherapy (CHART), in which a high dose of radiotherapy is given in a short time period. For small cell lung carcinoma cases that are potentially curable, chest radiation is often recommended in addition to chemotherapy The use of adjuvant thoracic radiotherapy following curative intent surgery for non-small cell lung carcinoma is not well established and is controversial. Benefits, if any, may only be limited to those in whom the tumor has spread to the mediastinal lymph nodes. For both non-small cell lung carcinoma and small cell lung carcinoma patients, smaller doses of radiation to the chest may be used for symptom control (palliative radiotherapy). Unlike other treatments, it is possible to deliver palliative radiotherapy without confirming the histological diagnosis of lung cancer,

sometimes for brain metastatic palliative radiotherapy 30Gy in 10 fraction used.
Brachytherapy (localized radiotherapy) may be given directly inside the airway when cancer affects a short section of bronchus. It is used when inoperable lung cancer causes blockage of a large airway Patients with limited stage small cell lung carcinoma are usually given prophylactic cranial irradiation (PCI). This is a type of radiotherapy to the brain, used to reduce the risk of metastasis. More recently, PCI has also been shown to be beneficial in those with extensive small cell lung cancer. In patients whose cancer has improved following a course of chemotherapy, PCI has been shown to reduce the cumulative risk of brain metastases within one year from 40.4% to 14.6%.
Recent improvements in targeting and imaging have led to the development of extracranial stereotactic radiation in the treatment of early-stage lung cancer. In this form of radiation therapy, very high doses are
Ch-35.indd 274 11-06-2015 10:26:11

delivered in a small number of sessions using stereotactic targeting techniques. Its use is primarily in patients who are not surgical candidates due to medical comorbidities.
Targeted therapy
In recent years, various molecular targeted therapies have been developed for the treatment of advanced lung cancer. Gefitinib (Iressa) is one such drug, which targets the tyrosine kinase domain of the epidermal growth factor receptor (EGFR), expressed in many cases of non-small cell lung carcinoma. It was not shown to increase survival, although females, Asians, nonsmokers, and those with bronchioloalveolar carcinoma appear to derive the most benefit from gefitinib.
Erlotinib (Tarceva), another tyrosine kinase inhibitor, has been shown to increase survival in lung cancer patients and has recently been approved by the FDA for second-line treatment of advanced non-small cell lung carcinoma. Similar to gefitinib, it also appeared to work best in females, Asians, nonsmokers, and those with bronchioloalveolar carcinoma.
The angiogenesis inhibitor bevacizumab, (in combination with paclitaxel and carboplatin), improves the survival of patients with advanced non-small cell lung carcinoma. However, this increases the risk of lung bleeding, particularly in patients with squamous cell carcinoma.
Advance sincytotoxic drugs, pharmacogenetics and targeted drug design] show promise. A number of targeted agents are at the early stages of clinical research, such as cyclo-oxygenase-2 inhibitors and the apoptosis promoter? Celecoxib, proteasome

inhibitors is bexarotene and the epidermal growth factor receptor inhibitor cetuximab,[] and vaccines.[] Future areas of research include ras proto-oncogene inhibition, phosphoinositide 3-kinase inhibition, histone deacetylase inhibition, and tumor suppressor gene replacement.

Prognosis: Non-small cell lung carcinoma staging and Manchester score

Prognostic factors in non-small cell lung cancer include presence or absence of pulmonary symptoms, tumor size, cell type (histology), degree of spread (stage) and metastases to multiple lymph nodes, and vascular invasion. For patients with inoperable disease, prognosis is adversely affected by poor performance status and weight loss of more than 10%. Prognostic factors in small-cell lung cancer include performance status, gender, stage of disease, and involvement of the central nervous system or liver at the time of diagnosis. For non-small cell lung carcinoma, prognosis is generally poor. Following complete surgical resection of stage IA disease, five-year survival is 67%. With stage IB disease, five-year survival is 57%. The five-year survival rate of patients with stage IV NSCLC is about 1%For small cell lung carcinoma, prognosis is also generally poor. The overall five-year survival for patients with SCLC is about 5%.Patients with extensive stage SCLC have an average five-year survival rate of less than 1%. The median survival time for limited-stage disease is 20 months, with a five – year survival rate of 20%.According to data provided by the National Cancer Institute, the median age of incidence of lung cancer is 70 years, and the median age of death by lung cancer is 71 years.

View publication stats